

Alternate definition. The function $z = f(x_1, x_2, \dots, x_n)$ is linear if there are multipliers $p_1, p_2, \dots, p_n$ for which

$$\Delta z = p_1 \cdot \Delta x_1 + p_2 \cdot \Delta x_2 + \dots + p_n \cdot \Delta x_n.$$